

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (first author), applied deep **transfer learning** to **neuroimaging / medical image classification**, a computer-vision approach to disease diagnosis.

Skills

Languages

Python, C++, JavaScript, TypeScript, SQL, Bash

ML / Deep Learning

PyTorch, JAX, NumPy, pandas, scikit-learn, CNNs / ResNet, transfer learning, image classification, diffusion, normalizing flows, autodiff

Evaluation & Experimentation

Model evaluation methodology, experimental design, statistical analysis (bootstrap CIs, McNemar’s test), anti-Goodharting / holdout protection, HuggingFace Transformers

Data & Pipelines

ML training-data pipelines, Apache Spark / PySpark, Spark UI, FastAPI, REST APIs, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **ML training-data pipeline / annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label clinical data for production models; cut manual annotation by **100+ hours/month** and improved labeling throughput by **60%**, accelerating model iteration in a healthcare setting.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Generative Models & Deep Learning (Image Models)

github.com/Deva-1903/UMASS_CICS_689

PyTorch, JAX, NumPy | UMass CS 689 Advanced ML

- Trained and benchmarked **5 deep architectures** (perceptron, MLP, VGG-style CNN, ResNet) across **3 optimizers** on **CIFAR-10 image classification**, tuning learning rates and documenting train/test-error curves (graded 87/100).
- Implemented **generative image models** from scratch in JAX, a normalizing flow with coupling layers and a **DDPM diffusion model**, plus a matrix-based reverse-mode **autodiff engine** validated against JAX to machine precision.

AutoEval - Model Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, ML Evaluation, Git Automation

- Designed an evaluation-improvement agent that proposes test cases and labels, scores them against multiple reference systems, and accepts only updates that improve **discriminative power** on a holdout-heavy fitness objective.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates to Git with report cards for an auditable, reproducible evaluation workflow.

Refusal Decay - Interpretability & Experimental Design

UMass CS 602 Research Project (graded full marks)

PyTorch, HuggingFace Transformers, Llama-3.1-8B-Instruct

- Ran a controlled interpretability study with multi-seed random/orthogonal controls and a benign positive control, reporting a **clean null** (0/300) with bootstrap **95% CIs** and McNemar’s test, and validating heuristic labels against a secondary classifier (87.5% agreement).

AgenticSearch - Multi-Stage ML Pipeline

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage ML pipeline (typed retrieval planning, extraction, verification) hardened with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation for reliable, debuggable production behavior.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.